Igor Lima Rocha Azevedo

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Education

Imperial College London — *London, UK | Sep 2025 - Sep 2026 (Expected)*

Master of Research (MRes) in Artificial Intelligence and Machine Learning

- Research Areas: Contrastive Deep Learning, Long-Tail Learning, Generative AI, Large Language Models.

University of Brasilia — *Brasilia, Brazil | Aug 2016 - Jun 2022*

Bachelor of Science in Electrical Engineering — First-Class Honours | GPA: 4.35 / 5.0

- Research Areas: FPGA Co-processors, Machine Learning for Audio Codecs, Deep Learning for Finance.

Relevant Experience

The University of Tokyo – Graph Neural Network Laboratory — *Tokyo, Japan*

Research Scholar | Advisors: Prof. Toyotaro Suzumura, Dr. Yuichiro Yasui — Apr 2023 – Apr 2025

- Designed novel news recommendation models with **Nikkei Inc.** (owner of the **Financial Times**).
- Built LLM systems including multi-agent pipelines, prompt engineering, and long-context handling with RAG.
- Trained large-scale distributed models on the Wisteria supercomputer.
- Implemented Point-of-Interest Graph Neural Network recommendation models in collaboration with **Toyota**.
- *Tools:* Python, PyTorch, JAX, RAG, LLM evaluation, distributed training.

VOGA – Technology Division, BTG Pactual (Latin America's Largest Bank) — *Brasilia, Brazil*

Intern → Developer → Development Team Lead — Jul 2021 – Apr 2023

- Delivered interactive data solutions and analytics for non-technical stakeholders throughout VOGA's acquisition by BTG Pactual, utilizing FastAPI/Plotly to build self-service dashboards integrated with BTG's API.
- Architected a centralized, real-time data ecosystem managing USD 300M+ in portfolios.
- *Tools:* Python, SQL, Snowflake, PostgreSQL, AWS, Terraform, Docker, Jenkins, FastAPI, Plotly, Git.

Cellcrypt Inc. – Advanced Security Systems Division — *London, UK (Remote)*

Research Intern | Advisors: Prof. Edson Mintsu Hung, Dr. Rogério Richa — Sep 2020 – Jun 2021

- Improved VoIP call quality by 7% in voice encrypted systems via automated ML parameter tuning.
- Analyzed voice quality metrics (MOS, jitter, latency) under varying network and encryption conditions.
- *Tools:* Python, PyTorch, Scikit-Learn, AWS (EC2, S3), Wireshark, VoIP Codecs (Opus, G.711).

Honors & Awards

3rd Place Winner, HackTheLaw King's College Cambridge — *Jun 2026*

University of Cambridge & Stanford CodeX

- Recognized for technical excellence and LLM integration by a panel of academic, industry (Perplexity, Anthropic), and law firms (Clifford Chance and White & Case).

Society for Industrial and Applied Mathematics (SIAM) Travel Award — *May 2025*

SDM'25 Conference

- Recognizing promising worldwide early-career researchers in data mining and machine learning research.

Japanese Government Research Scholarship (MEXT) — *Apr 2023 – Apr 2025*

Graduate Research Fellowship

- Approximately 9,000 students accepted out of 100,000 global applicants (∼9% acceptance rate).
- Full funding for tuition, living expenses, and research activities at The University of Tokyo.

Institutional Scientific Initiation Scholarship Program - PIBIC/CNPq — *Aug 2019 – Jul 2020*

Undergraduate Research Fellowship – Brazilian National Council for Scientific Development (CNPq)

- Awarded to the top ∼4% of undergraduates nationwide based on academic merit.
- Federal government-funded program for research in FPGA-based cryptographic security systems.

Publications

1. A Look Into News Avoidance Through AWRS: An Avoidance-Aware Recommender System — *2025*

Igor L.R. Azevedo, T. Suzumura, Y. Yasui

SIAM International Conference on Data Mining (SDM'25)

- Novel architecture for learning and modeling user avoidance patterns in news recommendation systems.

2. A SHA-3 Co-Processor for IoT Applications — *2020*

Igor L.R. Azevedo, A.S. Nery, A.C. Sena

IEEE Workshop on Communication Networks and Power Systems (WCNPS'20)

- FPGA-based implementation of SHA-3 cryptographic co-processor optimized for IoT security applications.